

NinaivX — Requirements Specification (MoSCoW)

Derived from the Literature Gap Analysis · Ashwin Kumar Ramesh Kumar · MSc AI, University of Sheffield

Requirements are specified using **MoSCoW** prioritisation (**Must**, **Should**, **Could**, **Won't have this time**) and are **derived from the gaps identified in the Literature Survey** (see the Gap Analysis). Each requirement records the **gap it addresses** and its **delivery status**, so the document shows why each requirement exists, what is delivered, and what is deliberately omitted. Requirements marked *Enabling* are engineering necessities that make the gap-driven requirements deliverable; those marked *Scope* come from the product vision rather than a literature gap.

Status key: ✓ Implemented · 🟡 Partial · ⚡ Planned · ✗ Omitted · — Out of scope

Gaps identified in the literature (source of the requirements)

ID	Gap in existing work	Evidence / systems	Ref
G1	Static memorials cannot hold a real conversation — they only replay recordings.	HereAfter AI	[16]
G2	Existing griefbots/companions are text-only, lacking natural voice interaction.	Project December, Replika, Xiaolce	[17][12–15] [11]
G3	Unconstrained generative models fabricate things the real person never said (hallucination).	GPT-3, Project December, YOV	[3][19][17] [26]
G4	No AI-transparency disclosure documented, although the ethics literature requires it.	YOV; Hollanek et al.	[25][26]
G5	Engagement-driven design fosters dependency; systems are not designed for grief and closure (one product frames grief as "a problem to be solved").	Replika, Xiaolce, YOV	[11][15][26]
G6	Sensitive data of the deceased is handled opaquely; consent and privacy unresolved.	Microsoft patent; US commercial products; Öhman & Floridi	[18][23][24]
G7	Systems are closed and proprietary; the architecture cannot be inspected or reproduced.	YOV, Microsoft patent	[18][26]
G8	No published evaluation of safety or effectiveness.	YOV, Project December	[17][26]
G9	Scope is limited to the deceased; no companion mode addressing loneliness.	YOV	[26]

MUST HAVE — core requirements without which the system has no value

ID	Requirement	Gap	Status	Where satisfied / notes
M1	Authenticate every request with a verified Supabase token (JWT).	G6	✓	app/core/auth.py; protects the deceased's sensitive data. Rejects invalid tokens (401).
M2	Onboard a user and store their profile (name, age, language).	Enabling	✓	POST /api/onboard, GET /api/me.

M3	Create personas of two types: deceased (legacy) and companion.	G9	✓	POST /api/personas; PersonaType enum. Extends scope beyond the deceased-only products.
M4	Route a conversation to the correct persona type automatically.	G9, G3	✓	Supervisor router; enables different safety profiles per mode.
M5	Hold a coherent, generative conversation in character as the persona.	G1	✓	POST /api/chat → LangGraph agent → Claude. Answers novel questions, unlike static memorials.
M6	Remember previous conversations across sessions.	G1	✓	Rolling summary checkpoint in Conversation table.
M7	Convert the AI reply to natural speech (text-to-speech).	G2	✓	voice_service.text_to_speech (ElevenLabs).
M8	Convert the user's speech to text (speech-to-text).	G2	✓	voice_service.speech_to_text (ElevenLabs Scribe).
M9	Support an end-to-end voice conversation (audio in → audio out).	G1, G2	✓	POST /api/voice-chat (full loop, verified live).
M10	Air-gap the deceased persona from the internet to prevent fabricated information.	G3, G6	✓	Deceased node has no tools bound — a code-level guarantee, not a prompt request.
M11	Enforce per-user data isolation; a user may only access their own personas.	G6	✓	Ownership check on every persona route.
M12	Persist users, personas and memory in a database.	Enabling	✓	Supabase (PostgreSQL) via SQLAlchemy.
M13	Ground the deceased persona only in its stored personality and memory (no invented sources).	G3	⚠	Grounded via prompt-context injection of personality + summary. A full vector-retrieval (RAG) pipeline over uploaded documents is not implemented — see C1.
M14	The architecture shall be openly documented and reproducible.	G7	✓	README, architecture diagram and full source; contrasts with closed commercial products.
M15	The safety design shall be empirically evaluated (project-level requirement).	G8	⌘	Planned: a technical evaluation measuring fabrication in the air-gapped legacy mode vs the web-connected companion mode, using synthetic personas . No human participants, so no ethical approval is required (no user study is conducted).

SHOULD HAVE — important, included but not strictly vital

ID	Requirement	Gap	Status	Where satisfied / notes
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S1	The companion persona may use the live internet for current information.	G9	✓	web_search tool bound to companion node (ReAct loop).
S2	Clone a voice from an uploaded audio sample.	G2	✓	POST /api/clone-voice (ElevenLabs).
S3	Assign a cloned or ready-made voice to a persona.	G2	✓	clone-voice (persona_id), select-voice, PATCH persona.
S4	Manage personas: list, view, update, delete.	Enabling	✓	GET/PATCH/DELETE /api/personas/{id}.
S5	Keep perceived latency low so conversation feels natural.	G1	✓	Memory summary runs as a FastAPI background task.
S6	Degrade gracefully if a voice provider key is absent (no crash).	Enabling	✓	Lazy voice service; returns 503 instead of failing at startup.
S7	Respond in the chosen language (per persona), colloquially.	Scope	✓	Each persona has its own language ; the prompt drives natural, colloquial, slang-aware replies and TTS speaks them. Native <i>accent</i> depends on the voice asset (upload/clone a native voice — the default library is English).
S8	Produce speech-appropriate text (no emojis or stage directions).	G2	✓	TTS-tuned system prompts (agents/prompts.py).
S9	Disclose to the user that they are speaking with an AI (transparency).	G4	✓	Persona stays in character but honestly confirms it is an AI if directly asked (verified live); public notice via GET /api/disclosure and in profile/persona responses.
S10	Support healthy grieving and closure; avoid engagement-maximising or dependency-forming design.	G5	ⓘ	By design: prompts orient toward comfort and closure; no gamification, streaks or engagement mechanics. Effectiveness to be assessed in the evaluation (M15).

COULD HAVE — desirable; omitted for now, candidates for future work

ID	Requirement	Gap	Status	Notes
C1	True retrieval-augmented generation (RAG) over uploaded memories/documents via a vector store.	G3	X	Omitted. Would deepen the grounding in M13; current grounding is prompt-context only. Clear future work.
C2	Real-time streaming / full-duplex audio (sub-second response).	G1, G2	X	Omitted. Current design is request/response (turn-based), not streaming.
C3	Meter voice usage (minutes) to support freemium limits.	Scope	X	Omitted. Commercial concern; not required for the research prototype.
C4	Production hardening: rate limiting and restricted CORS.	Enabling	X	Omitted. CORS is currently open ("*") for development.
C5	Store full conversation transcripts (not only summaries).	G6	X	Deliberately omitted for privacy; memory is summary-based by design.

WON'T HAVE (this time) — explicitly out of scope for this dissertation

ID	Requirement	Gap	Status	Notes
W1	Real-time video / lip-synced avatars of the persona.	Scope	—	Future vision (roadmap Phase 3); out of scope.
W2	Dedicated multilingual ("Polyglot") models for ancestral languages.	Scope	—	Future expansion; beyond current timeframe.
W3	Payment, subscription and billing integration.	Scope	—	Commercial concern, not part of the research artefact.
W4	Public mobile app-store deployment.	Scope	—	The dissertation delivers the backend + evaluation, not a shipped app.
W5	Company incorporation, legal and GDPR consultancy.	Scope	—	Outside the technical scope of the dissertation.

Traceability — every identified gap is addressed by at least one requirement

Gap	Addressed by	Coverage status
G1	M5, M6, M9, S5 (C2 future)	✓ Delivered
G2	M7, M8, M9, S2, S3, S8 (C2 future)	✓ Delivered
G3	M4, M10, M13 (C1 future)	✓ Air-gap delivered; ➊ grounding basic
G4	S9	✓ Delivered
G5	S10 (assessed via M15)	➊ By design; evaluation pending
G6	M1, M10, M11, C5	✓ Delivered
G7	M14	✓ Delivered
G8	M15	⌘ Planned (dissertation evaluation)
G9	M3, M4, S1	✓ Delivered

Summary: 15 Must-have requirements (13 implemented, 1 partial — M13 grounding, 1 planned — M15 evaluation); 10 Should-have (9 implemented, 1 partial — S10 grief-appropriate design); 5 Could-have recorded as future work; 5 Won't-have explicitly out of scope. All nine literature gaps (G1–G9) are traceable to at least one requirement.